

BUDYONNOVSK, Russian FederationWMO: **370610**Lat: **44.783N**Lon: **44.133E**Elev: **136**StdP: **99.70**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-13.9	-20.7	0.6	-17.7	-16.5	0.9	-13.6	8.6	0.8	7.1	-0.6	1.4	80	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	36.3	20.3	34.6	19.9	32.9	19.6	22.3	31.0	21.5	30.2	20.8	29.3	3.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.9	25.0	19.0	14.0	24.3	18.1	13.3	23.9	66.3	31.0	63.2	30.1	60.8	29.4	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.3	5.3	-22.0	38.5	4.9	2.0	-25.5	40.0	-28.4	41.2	-31.1	42.4	-34.7	43.8		
(5)			-22.2	24.4	4.9	1.6	-25.7	25.5	-28.6	26.4	-31.3	27.3	-34.8	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-2.5	-1.2	4.4	11.1	17.5	22.5	25.6	25.1	18.8	11.8	4.6	0.2		
	DBStd	10.79	5.58	5.80	3.99	4.01	3.92	3.40	2.87	3.26	3.90	4.81	4.38	4.65		
	HDD10.0	1417	387	313	177	33	2	0	0	0	1	36	166	303		
	HDD18.3	3132	645	547	432	219	63	7	0	1	40	206	412	561		
	CDD10.0	1993	0	0	4	66	234	376	485	467	265	93	4	0		
	CDD18.3	665	0	0	0	2	37	132	227	209	54	4	0	0		
	CDH23.3	7280	0	0	1	30	387	1412	2622	2350	446	33	0	0		
	CDH26.7	3337	0	0	0	4	101	610	1336	1157	124	4	0	0		
(14) Wind	WSAvg	2.3	2.2	2.4	2.8	2.6	2.3	2.3	2.2	2.2	2.1	2.1	2.2	2.3		
(15) Precipitation	PrecAvg	439	27	23	32	40	50	56	38	42	35	36	30	30		
	PrecMax	574	59	44	73	109	91	101	85	102	102	82	66	85		
	PrecMin	279	3	9	9	3	6	11	5	2	5	2	4	14		
	PrecStd	59	14	9	17	25	22	24	23	30	27	23	16	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	13.0	19.7	26.5	31.7	36.5	38.2	38.2	32.9	27.1	16.9	10.9		
		MCWB	6.0	8.3	10.8	15.3	18.4	20.3	20.5	20.3	20.8	16.5	12.5	8.3		
	2%	DB	6.1	9.0	15.7	23.5	29.3	34.0	36.5	36.2	29.6	23.3	13.8	8.6		
		MCWB	4.4	6.0	9.3	14.2	17.4	19.5	20.2	20.5	18.7	15.7	11.1	6.9		
	5%	DB	4.6	6.4	12.9	21.0	27.5	32.1	34.8	34.4	27.7	20.6	11.7	7.1		
		MCWB	3.6	4.3	7.9	13.2	17.0	19.1	19.7	20.2	18.2	14.9	9.7	5.9		
	10%	DB	3.4	4.9	10.7	18.7	25.5	30.2	33.0	32.5	25.9	18.4	10.3	5.7		
		MCWB	2.5	3.6	6.8	12.1	16.4	18.5	19.4	19.9	17.4	14.2	8.8	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	8.4	12.0	16.4	20.2	22.3	23.4	23.7	22.5	18.0	13.1	8.7		
		MCDB	8.4	12.2	17.8	24.3	28.3	32.4	32.8	31.9	29.6	23.1	15.9	10.1		
	2%	WB	4.7	6.2	9.9	15.0	18.8	21.1	22.1	22.5	20.4	16.7	11.5	7.1		
		MCDB	5.8	8.5	14.6	22.2	26.3	30.5	31.2	30.7	26.9	21.1	13.2	8.3		
	5%	WB	3.7	4.7	8.5	13.8	18.0	20.2	21.2	21.7	19.2	15.7	10.1	6.0		
		MCDB	4.6	6.0	12.2	19.8	25.2	28.8	30.0	30.1	25.7	19.6	11.4	6.9		
	10%	WB	2.6	3.6	7.1	12.7	17.1	19.5	20.6	21.0	18.2	14.7	8.9	4.9		
		MCDB	3.3	4.8	10.3	17.7	23.6	27.6	29.1	29.4	24.0	17.9	10.1	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.6	8.4	10.8	11.3	12.0	12.1	12.2	11.1	8.7	6.2	5.2		
		MCDBR	7.0	9.2	13.7	14.6	14.7	14.9	15.1	15.0	13.7	12.3	9.4	7.1		
	5% WB	MCWBR	5.2	6.4	7.7	6.6	5.4	4.6	4.0	4.5	4.8	5.8	6.3	5.4		
		MCWBR	6.2	8.5	12.3	12.8	12.6	13.1	13.0	12.9	12.0	10.6	7.9	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.380	0.418	0.470	0.437	0.442	0.466	0.472	0.441	0.395	0.362	0.352		
		taud	2.232	2.156	2.107	2.000	2.180	2.218	2.156	2.129	2.175	2.264	2.295	2.226		
	Ebn at Noon		739	779	802	792	836	832	804	783	773	757	718	684		
		Edh at Noon	88	114	138	166	143	139	146	145	128	102	83	81		
(44) All-Sky Solar Radiation	RadAvg		1.11	1.94	2.94	4.62	5.78	6.52	6.41	5.74	4.19	2.45	1.30	0.90		
	RadStd		0.31	0.44	0.34	0.49	0.56	0.35	0.38	0.29	0.40	0.31	0.22	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page